

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. What kind of state-space model should I use if I wish to track a target that is nearly motionless?

1 / 1 point

- ☒ A "nearly constant position" (NCP) model.
- ☐ A "coordinated turn" (CT) model.
- ☐ A "nearly constant velocity" (NCV) model.
- ☐ A "nearly constant acceleration" (NCA) model.

✔ Correct
Yes, this is correct.

2. What makes the target-tracking application different from other applications of the Kalman filter? (Select all that apply.)

1 / 1 point

- ☐ In the target-tracking application, we do not know $v(t)$.
- ☒ In the target-tracking application, we do not know $u(t)$.

✔ Correct
Yes, this is correct.

- ☒ In the target-tracking application, we do not know precise values of the state-space-model A , B , C , and D matrices that describe the object being tracked.

✔ Correct
Yes, this is correct. This is why we use approximate models such as NCP, NCV, and CT.

- ☐ In the target-tracking application, we do not know $w(t)$.

3. Run the Coursera Labs Jupyter notebook for this lesson titled "Jupyter notebook used in conjunction with practice quiz." This notebook implements simulations of different models commonly used in target-tracking applications.

1 / 1 point

Do not modify the code line "randn("state",0)". This operation forces the pseudo-random number generator in Octave to generate a reproducible sequence of random values, which is needed for the notebook to generate answers that can be verified by the quiz grader.

For this question, simply execute the lab and enter the final x -coordinate value (the ξ variable) for the NCV simulation. Enter three digits after the decimal point.

-0.067324

✔ Correct
Yes, this is correct.